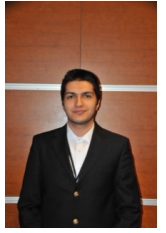


Burak Ege Kaya

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github.com/burakkeynz | Kagithane / Istanbul



EDUCATION

Bahcesehir University, Istanbul

BSc in Computer Engineering (4th Year)

GPA: 3.14/4.0

Expected July 2026

BSc in Management Engineering (Double Major, 3rd Year)

GPA: 3.43/4.0

Expected Jun 2027

WORK EXPERIENCE

AI Software Developer Intern, Baykar Technologies

June 2026 – July 2026 (Upcoming)

- Accepted as an AI Software Developer for Baykar Technologies' 2026 Summer Internship Program before my graduation.

AI Trainee, Yapay Zeka ve Teknoloji Akademisi

Dec 2025 – Present

- Selected as one of 1,500 participants among 31,700 applicants for a project-based national program supported by the Ministry of Industry and Technology, Google Türkiye, and T3 Foundation.
- Collaborating on AI-driven initiatives within a competitive ecosystem focusing on large-scale tech integration.

AI Trainee, Ministry of Industry and Technology (Milli Teknoloji Akademisi)

Jan 2026 – Present

- Admitted to the AI Specialization Program with limited candidates (120) after ranking among the top performers of 1000 candidates who had been selected over 8000+ applicants in intensive foundational training (Data Structures, OS, DBMS).
- Focusing on Advanced Python, Cloud Systems, GPU Accelerators, Parallel Programming and other AI specialization topics.
- Will be developing a capstone project under industry mentorship to graduate from program.

Software Development Intern, Cizgi Teknoloji Elektronik Tasarım ve Üretim

Jul 2025 – Aug 2025

- Developed a full-fledged hospital communication panel web app under senior R&D mentorship.
- Built real-time communication features using React, FastAPI, WebRTC, and Socket.IO.
- Integrated an AI assistant (Gemini 2.5) with live audio transcription using AudioWorklet.
- Delivered a production-ready demo for real-time interaction between medical staff.

Data Analyst Intern, Kaan Havalı Kırıcı Makina

Jul 2024 – Sep 2024

- Optimized Python/Django backend logic for faster data handling and processing.
- Generated visual analytics dashboards with Pandas, NumPy and Matplotlib to monitor production efficiency. Used Watchdog library to track file changes and update database.

PROJECTS

AgentBench-TR: Built a 4-agent LangGraph pipeline with hybrid BM25 and ChromaDB retrieval using Reciprocal Rank Fusion. Implemented claim-level hallucination detection in the ValidatorAgent and an automatic retry loop on low-confidence answers. Designed an evaluation layer tracking consistency, hallucination rate, cost, and latency per trace, exposed through a FastAPI backend and a Plotly Dash dashboard. [\[GitHub\]](#)

AeroGuard: Built an end-to-end ML pipeline on the NASA CMAPSS dataset to predict Remaining Useful Life of turbofan engines. Engineered temporal features over 5–20 cycle windows, benchmarked five models, and achieved RMSE of 1.50 with Random Forest. Applied SHAP TreeExplainer for global and local interpretability and built a three-tier early warning system. [\[GitHub\]](#)

Autonomous Driving via Deep Reinforcement Learning (highway-env): Developed as a term project in collaboration with a teammate using Google Colab. Implemented DQN and SAC algorithms to solve complex autonomous driving scenarios including Highway, Merge, Roundabout, and Racetrack environments. Optimized reward functions and trained agents using Stable Baselines3 to achieve high-speed navigation and strategic merging while maintaining a zero-collision policy. [\[GitHub\]](#)

Hospital Panel System: Under mentorship, built a real-time hospital communication platform as the full-stack developer responsible for all core modules. Developed with React/Redux and FastAPI using Socket.IO for real-time data exchange and WebRTC for audio/video calls. Integrated AI assistance powered by Gemini 2.5, implemented live speech-to-text via AudioWorklet (48→16 kHz downsampling, VAD), and added Cookie-based JWT authentication, AWS S3 file storage, and encrypted PDF exports for secure workflows. [\[GitHub\]](#)

Chovora Web Platform: Leveraging a Computer Engineering background within a Management Engineering marketing course, took full ownership of all technical development while teammates focused on marketing strategy. Implemented JWT-based authentication, dynamic cart features, an admin dashboard, and deployed the production app on Render and Vercel. Collaborated with non-technical teammates on UX decisions and iterated based on their feedback. chovora.com

SKILLS & TECHNOLOGIES

Soft Skills: Project Management, Leadership, Strategic Thinking, Communication, Agile Development

Programming Languages: Python, JavaScript, C#, Java

AI & Data: NumPy, Pandas, Matplotlib, Scikit-learn, XGBoost, SHAP, Stable Baselines3, HuggingFace Transformers, LangChain, ChromaDB, Ollama, Gemini API, OpenAI API

Frontend: React, HTML, CSS

Backend & Database: .NET, FastAPI, Node.js, MySQL, Oracle DB, MongoDB

Realtime Communication: Socket.IO, WebRTC

DevOps & Tools: Docker (MySQL), Git, AWS S3, Google Colab, VS Code, Excel

LANGUAGES

Turkish: Native

English: Professional (C1)

German: Beginner (A1)

CLUBS & VOLUNTEER WORK

Core Organization Team Member, BAU Management Engineering Club

Oct 2024 - Apr 2026

Actively took part in the organizational unit of the university's career-oriented club alongside management engineering students. As an organization team member, I was responsible for planning and coordinating operational processes such as selecting guest speakers, setting the event calendar, preparing interview questions, structuring the event flow, and organizing audience seating. Contributed to ensuring that all organizational aspects of the events ran smoothly and professionally with my teammates.

CERTIFICATES

McKinsey.org Forward Program - McKinsey & Company, Dec 2025

Mastering WebRTC – Robert Bunch (Udemy), Jul 2025

FastAPI – The Complete Course 2025 (Eric Roby, Udemy), Jun 2025

React Basics – Meta (Coursera), Jun 2025

The Complete JavaScript Course 2025 – Jonas Schmedtmann (Udemy), Feb 2025

Programming with JavaScript – Meta (Coursera), Apr 2025

Excel Skills for Business – Macquarie University (Coursera), Dec 2024